

EXPERIMENT 1

Setting Up the Environment and Data Preprocess

Import Libraries:

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[7]: data = pd.read_csv("Datasets/students.csv")
```

```
[8]: data
```

```
[8]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.0	F
1	102	Ravi	21	76.0	M
2	103	John	20	NaN	M
3	104	Meena	22	91.0	F
4	105	Kiran	21	88.0	M
5	106	Priya	20	95.0	F
6	106	Priya	20	95.0	F

Display Dataset:

```
[8]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.0	F
1	102	Ravi	21	76.0	M
2	103	John	20	NaN	M
3	104	Meena	22	91.0	F
4	105	Kiran	21	88.0	M
5	106	Priya	20	95.0	F
6	106	Priya	20	95.0	F

```
[9]: data.head()
```

```
[9]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.0	F
1	102	Ravi	21	76.0	M
2	103	John	20	NaN	M
3	104	Meena	22	91.0	F
4	105	Kiran	21	88.0	M

```
[10]: data.head()
```

```
4      105      Kiran      21      88.0      M
```

```
[11]: data.tail()
```

```
[11]:
```

	RollNo	Name	Age	Marks	Gender
2	103	John	20	NaN	M
3	104	Meena	22	91.0	F
4	105	Kiran	21	88.0	M
5	106	Priya	20	95.0	F
6	106	Priya	20	95.0	F

Dataset Shape:

```
[12]: data.shape
```

```
[12]: (7, 5)
```

```
[12]: data.columns
```

Column Names:

```
[13]: data.columns
```

```
[13]: Index(['RollNo', 'Name', 'Age', 'Marks', 'Gender'], dtype='str')
```

Data Types:

```
[13]: Index(['RollNo', 'Name', 'Age', 'Marks', 'Gender'], dtype='str')
```

```
[14]: data.dtypes
```

```
[14]: RollNo      int64  
      Name      str  
      Age      int64  
      Marks  float64  
      Gender    str  
      dtype: object
```

Dataset Information:

```
[1]: data.info()

<class 'pandas.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 5 columns):
#   Column  Non-Null Count  Dtype  
---  -
0   RollNo  7 non-null      int64   
1   Name     7 non-null      str      
2   Age      7 non-null      int64   
3   Marks    6 non-null      float64  
4   Gender   7 non-null      str      
dtypes: float64(1), int64(2), str(2)
memory usage: 412.0 bytes
```

Summary Statistics:

```
[16]: data.describe()

[16]:
```

	RollNo	Age	Marks
count	7.000000	7.000000	6.000000
mean	103.857143	20.571429	88.333333
std	1.951800	0.786796	7.201852
min	101.000000	20.000000	76.000000
25%	102.500000	20.000000	85.750000
50%	104.000000	20.000000	89.500000
75%	105.500000	21.000000	94.000000
max	106.000000	22.000000	95.000000

Missing Values:

```
[17]: data.isnull().sum()

[17]: RollNo      0
      Name       0
      Age       0
      Marks     1
      Gender    0
      dtype: int64
```

Filled Missing Values:

```
[18]: data["Marks"] = data["Marks"].fillna(data["Marks"].mean())
```

```
[19]: data
```

```
[19]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.000000	F
1	102	Ravi	21	76.000000	M
2	103	John	20	88.333333	M
3	104	Meena	22	91.000000	F
4	105	Kiran	21	88.000000	M
5	106	Priya	20	95.000000	F
6	106	Priya	20	95.000000	F

Duplicate Records:

```
[20]: data.duplicated().sum()
```

```
[20]: np.int64(1)
```

Duplicate Removal:

```
[30]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.000000	0
1	102	Ravi	21	76.000000	1
2	103	John	20	88.333333	1
3	104	Meena	22	91.000000	0
4	105	Kiran	21	88.000000	1
5	106	Priya	20	95.000000	0

```
[ ]:
```

```
[31]: data.drop_duplicates(inplace=True)
```

```
[32]: data
```

```
[32]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.000000	0
1	102	Ravi	21	76.000000	1
2	103	John	20	88.333333	1
3	104	Meena	22	91.000000	0
4	105	Kiran	21	88.000000	1
5	106	Priya	20	95.000000	0

Label Encoding:

```
[33]: from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

data["Gender"] = le.fit_transform(data["Gender"])
```

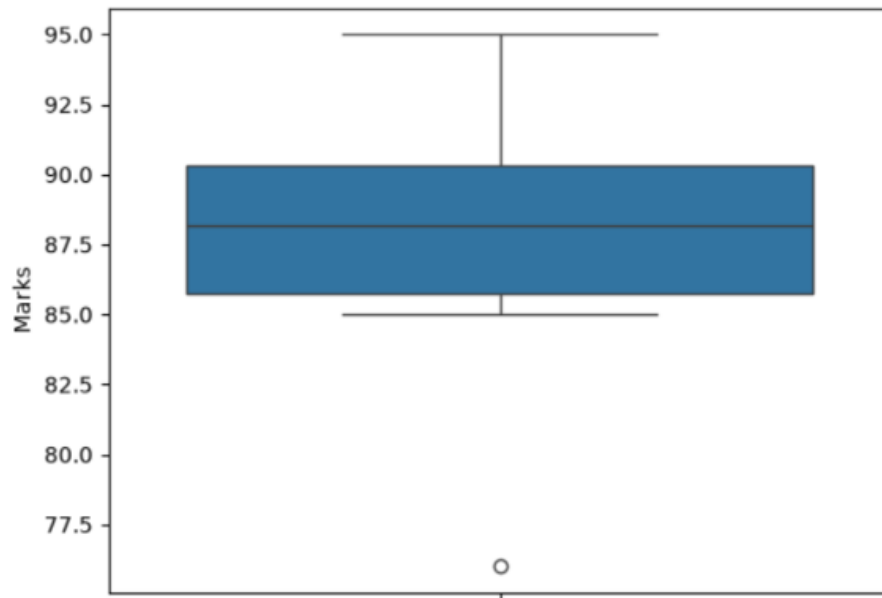
```
[34]: data
```

```
[34]:
```

	RollNo	Name	Age	Marks	Gender
0	101	Asha	20	85.000000	0
1	102	Ravi	21	76.000000	1
2	103	John	20	88.333333	1
3	104	Meena	22	91.000000	0
4	105	Kiran	21	88.000000	1
5	106	Priya	20	95.000000	0

Box Plot:

```
[35]: sns.boxplot(y=data["Marks"])  
plt.show()
```



One-Hot Encoding:

```
[36]: colors = pd.DataFrame({  
    "Color": ["Red", "Blue", "Green", "Red"]  
})  
  
pd.get_dummies(colors)
```

```
[36]:
```

	Color_Blue	Color_Green	Color_Red
0	False	False	True
1	True	False	False
2	False	True	False
3	False	False	True

Standardization:

```
[37]: from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled = scaler.fit_transform(data[["Age", "Marks"]])

scaled

[37]: array([[ -0.89442719, -0.377897  ],
 [  0.4472136 , -1.90837984],
 [ -0.89442719,  0.1889485  ],
 [  1.78885438,  0.6424249  ],
 [  0.4472136 ,  0.13226395],
 [ -0.89442719,  1.32263949]])
```

Normalization:

```
[38]: from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled = scaler.fit_transform(data[["Age", "Marks"]])

scaled

[38]: array([[ -0.89442719, -0.377897  ],
 [  0.4472136 , -1.90837984],
 [ -0.89442719,  0.1889485  ],
 [  1.78885438,  0.6424249  ],
 [  0.4472136 ,  0.13226395],
 [ -0.89442719,  1.32263949]])
```